

UNIVERSITI TEKNOLOGI MARA

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES
COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATICS

CSC649 - SPECIAL TOPICS IN COMPUTER SCIENCE

TITLE:

INDIVIDUAL ASSIGNMENT

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PROGRAMME	CS230
GROUP	5D

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1. 0 Introduction

Supercook is a popular cook apps recipe discovery platform designed to help users find meals based on the ingredients they currently have at home. By allowing users to input available items, the app generates matching recipes to maximize food utility and simplify meal planning. The app uses Natural Language Processing techniques by choosing specific ingredients to determine the meals based on what the user has. Therefore, several issues have to be solved where users may still struggle to decide how to turn leftover food into new meals. If this problem is not going to be solved, it can lead to food waste.

2.0 Problem formulation

2.1 Problem Statement

Home cooks using recipe applications to prepare meals from available ingredients may encounter missing pantry information, disappearing saved recipes, and recommendations containing unavailable ingredients. For example, Bill S (2023) reported that ingredients entered on one computer did not appear on another computer using the same account, while DJ (2024) reported that saved recipes disappeared and the application reset when switching between apps. ReNew Innovations (2026) also reported that SuperCook suggested recipes using out-of-stock pantry items. These reported problems interrupt meal planning and make it harder for users to identify recipes they can prepare with their current ingredients. Therefore, this project proposes an NLP-assisted recipe recommendation system with ingredient-availability tracking and persistent pantry storage to address these reported difficulties.

2.2 Users and Stakeholders

1. Primary Users: Individuals looking to cook meals using available pantry leftovers, reduce grocery bills, and minimize household food waste.
2. Secondary Users: Users who require immediate recipe discovery without forced account setup.
3. Stakeholders: System developers and UI/UX designers aiming to deliver seamless data synchronization and effective recipe matching algorithms.

2.3 Project Scope

In-Scope:

1. Implementation of an offline-first local storage system with background cloud synchronization for saved recipes and pantry lists.
2. Integration of Natural Language Processing (NLP) text processing for dynamic ingredient-recipe matching
3. Development of an ingredient availability tracking system (In-Stock / Out-of-Stock toggles)
4. Introduction of context-aware ingredient substitution suggestions when items are missing

Out-of-Scope:

1. Direct API integration with commercial grocery ordering platforms (e.g., Lotus's, GrabMart).
2. Paid subscription and payment gateway processing.

3.0 Requirements

3.1 Functional requirement

ID	Functional requirement	Testability Method
FR1	The system shall allow users to add an ingredients from their pantry	Verified by entering the ingredient name and making sure it appears in the pantry list.
FR2	The system shall allow users to mark each pantry ingredient as available or out of stock.	Verified by clicking the stock toggle and confirming the visual availability state updates.
FR3	The system shall exclude out-of-stock pantry ingredients when suggesting new recipe based on user's search	Verified by marking an item out-of-stock and running a search to ensure recipes requiring that item are excluded.
FR4	The system shall display substitution suggestions when an approved rule exists for the selected missing ingredients and recipe context.	Verified by selecting a recipe with a missing ingredient and confirming valid substitutes are displayed.
FR5	The system shall list the missing ingredients for each displayed recipe.	Verified by opening a recipe detail view and checking the highlighted missing

		ingredients list.
FR6	The system shall allow users to save a recipe to their favourites.	Verified by clicking the favourite icon and confirming the recipe appears in the Favourites tab.
FR 7	The system shall restore locally saved pantry data when the application is reopened.	Verified by force-closing the app and confirming all pantry entries persist upon relaunch.
FR 8	The system shall synchronize a signed-in user's pantry changes across devices when an internet connection is available.	Verified by editing pantry items on Device A and confirming automatic sync on Device B.
FR 9	The system shall display whether pantry changes are saved locally, pending synchronization, or synchronized.	Verified by observing status badges across offline, syncing, and synchronized states.
FR 10	The system shall allow users to search for recipes without creating an account.	Verified by launching the app in Guest Mode and executing a recipe search without logging in.

3.2 Non-Functional Requirements

ID	Non-functional Requirement
NFR 1	Performance - for a dataset of 1000 recipes containing up to 30 ingredients, the system needs to display the result within 5 seconds.
NFR 2	Usability - At least 4 of 5 first-time test participants shall add three ingredients and open a matching recipe within 2 minutes without assistance.
NFR 3	Reliability - the message "saved locally" confirmation, whenever system restore all pantry entries and their availability states without loss
NFR 4	Synchronization - with both devices online and no simultaneous edits, a confirmed pantry change shall appear on the second signed-in device within 10 seconds in at least 19 of 20 trials.

3.3 User Stories and Acceptance Criteria

ID	User Story	Acceptance Criteria
US1 (FR8, FR9)	As a home cook, I want my pantry list changes to synchronize automatically across my mobile and laptop, so that I can manage my meal plans seamlessly without losing data when switching devices.	● Editing a pantry item on one logged-in device updates the cloud database and reflects on the second device within 10 seconds. ● When offline, the app displays a "Saved Locally" indicator and pushes updates automatically once reconnected.
US2 (FR2, FR3)	As a grocery planner, I want to toggle missing pantry items as out-of-stock, so that the app only suggests recipes I can actually prepare without buying extra items.	● Toggling an ingredient to "Out-of-Stock" immediately excludes recipes requiring that item from current search recommendations. ● Re-toggling the ingredient to "Available" restores relevant recipe recommendations automatically.
US3 (FR10)	As a first-time visitor, I want to search for recipes immediately in Guest Mode without registering an account, so that I can test the app's features quickly without sign-up friction.	● Clicking "Continue as Guest" bypasses the login screen and navigates directly to the recipe search interface. ● Guest users can input ingredients and view matching recipe lists without being blocked by forced registration popups.

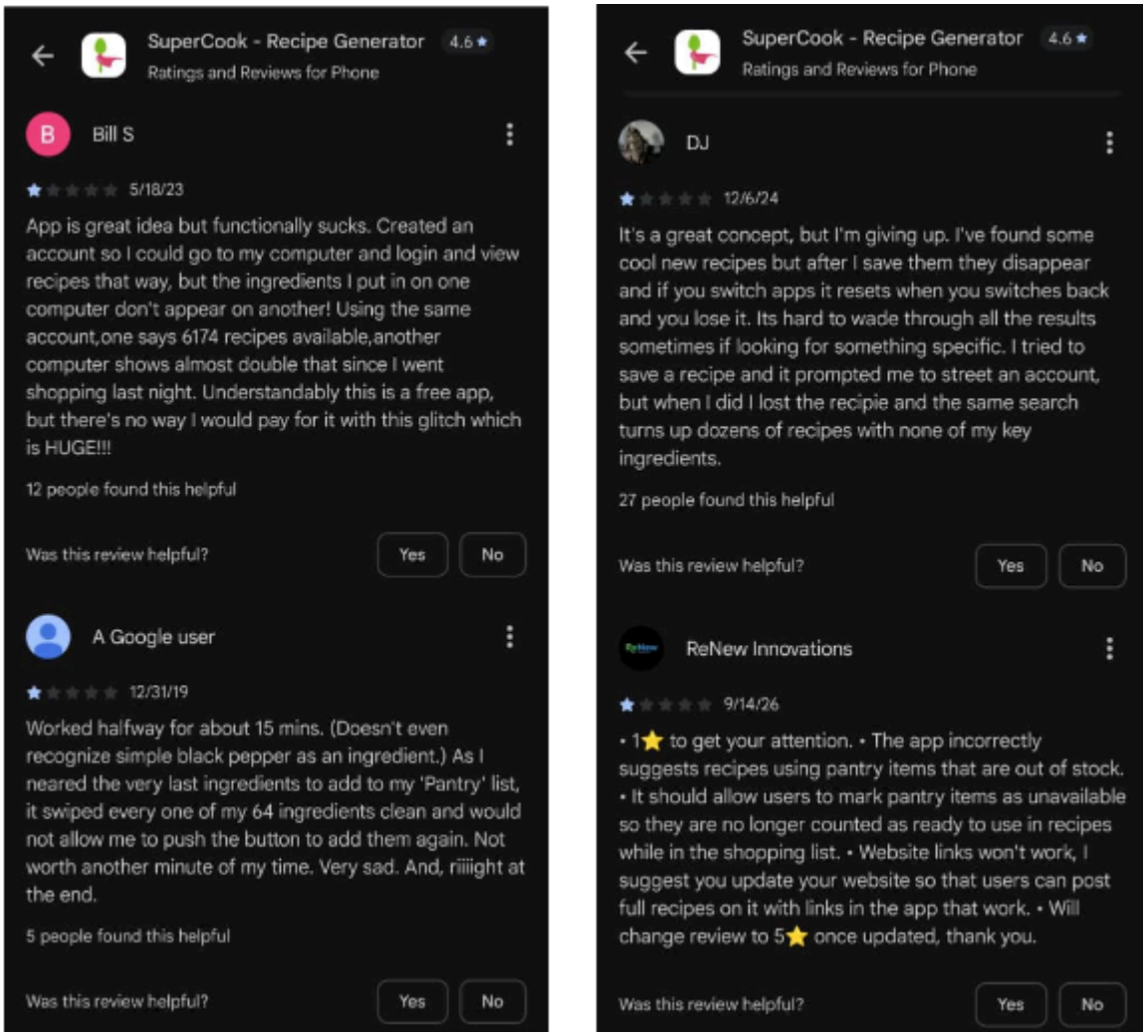


Figure 3.0 User's review on Supercook app from Google App Store

4.0 Specialized technology

4.1 Fundamental Concepts

The specialized technology that I used in this project is Natural Language Processing (NLP). The system integrated with Content-Based Filtering and Offline-First Synchronization Architecture. This is because the supercook apps process ingredient description by users and retrieve relevant recipes based on the user's search. Traditional software having a rigid SQL query execution where required to exact string matches. For example, matching the exact term "chicken" with "chicken". When home cooks input arbitrary or non-standard leftover ingredient strings (e.g., "200g organic fresh chicken breast" or "diced tomatoes") these static database filters fail to match recipe items, resulting in a search failure. (Peng, 2024).

To resolve these limitations, modern meal planning platforms integrate Natural Language Processing (NLP) and Content-Based Filtering within a Vector Space Model (VSM). Instead of performing Boolean string matching, NLP pre-processing pipelines transform unstructured ingredient text lists into continuous, high-dimensional numerical weight matrices. The recommendation engine calculates the mathematical and semantic proximity between the vector representation of the user’s available pantry inventory and candidate recipe vectors in the system repository to deliver dynamic, personalized meal suggestions

To convert raw user ingredient strings into structured numerical vector representations, the system executes a four-stage processing pipeline.

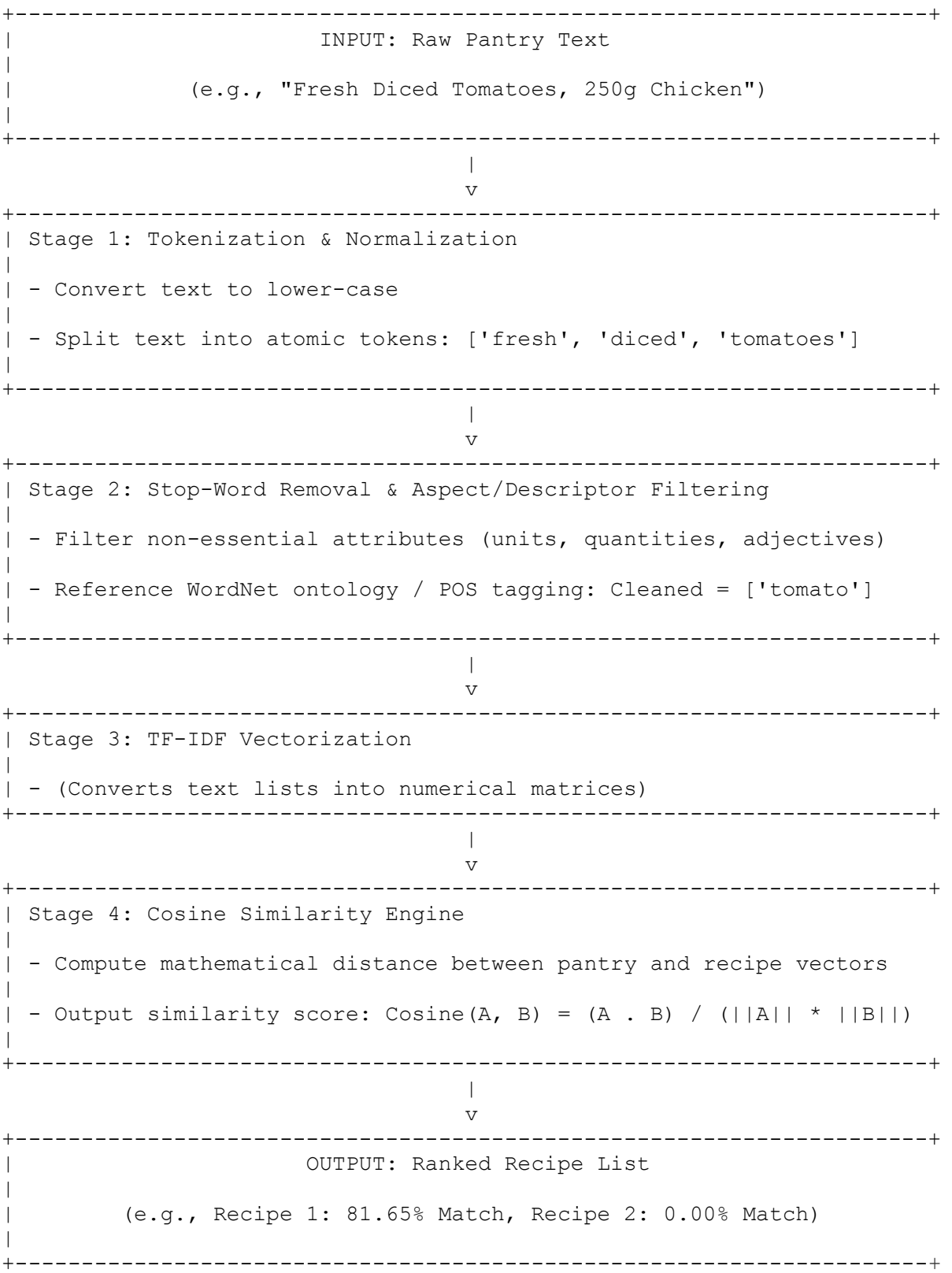


Figure 4.0: Four-Stage NLP Ingredient Vectorization and Recipe Retrieval Pipeline

1. **Tokenization and Normalization**

   While inputting the ingredients text it will be converted into lower-case characters and suited with standard tokenization rules. For example: (*"Fresh Diced Tomatoes"* → “Fresh”,”Diced”,*"tomato"*)
2. **Stop-Word Removal & Aspect Extraction**

   Filtering out measurement units, quantities, and non-essential descriptors. For example: ("grams", "chopped", "fresh").
3. **TF-IDF Vectorization**

   Converting cleaned text lists into numerical weight matrices using Term Frequency-Inverse Document Frequency (TF-IDF).
4. **Cosine Similarity Computation**

   Calculating the mathematical distance between the user's available pantry vector and candidate recipe vectors to output ranked similarity percentages.

$$\text{Cosine Similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

4.2 Framework, Models, Methods and Tools

1. TF-IDF (Term Frequency-Inverse Document Frequency) Model

- Theoretical Principle & Author Context: As demonstrated by Pal (2023), TF-IDF is a statistical NLP vectorization technique used to convert unstructured text items into numerical weight matrices. The model calculates term frequency while penalizing universally common terms across the dataset corpus.
- Application in System: In this system, ubiquitous ingredients (e.g., "water", "salt") receive lower mathematical weights, whereas distinctive leftover ingredients (e.g., "parmesan", "salmon") receive higher weights. This prevents common staples from skewing recipe match scores.

2. Cosine Similarity Algorithm

- Theoretical Principle & Author Context: According to Pal (2023) and Deac-Petruşel & Limboi (2020), Cosine Similarity is a spatial metric that measures the cosine of the angle between two multi-dimensional vectors in a Vector Space Model (VSM)
- Application in System: The algorithm evaluates how closely a user's active pantry inventory vector matches candidate recipe vectors stored in the database. It enables flexible, fuzzy, and partial matching when ingredient descriptors contain non-standard phrasing or when secondary ingredients are missing

.

3. Scikit-Learn (Python Machine Learning Framework)

- Theoretical Principle & Author Context: Pal (2023) highlights Scikit-Learn as an open-source Python machine learning library providing efficient data structures for text pre-processing, matrix operations, and content-based recommendation algorithms.
- Application in System: Scikit-Learn is used to structure the automated vectorization pipeline (TfidfVectorizer), compute high-dimensional distance matrices, and execute similarity search operations rapidly during user query processing

.

4. Hybrid Offline-First State Synchronization Engine

- Theoretical Principle & Author Context: Benchmark research by Churik (2025) establishes that a hybrid synchronization architecture—combining local persistent storage (SQLite/IndexedDB) with optimistic UI updates and Conflict-Free Replicated Data Types (CRDT)—guarantees data availability under intermittent network connectivity.
- Application in System: All pantry updates and saved recipes are written instantly to local storage with optimistic rendering, while background REST/WebSocket push-pull protocols synchronize deltas with the server. This architecture maintains over 98% cross-device data consistency and prevents data loss or state resets when users switch applications or lose connection.

4.3 Fit to Problem, Benefits, Limitations, and Alternative Approach:

Fit to Problem

Integrating NLP vector retrieval with an offline-first state synchronizer directly resolves the reported failures in Supercook. Rather than rejecting queries when exact ingredients are missing, the NLP engine ranks recipes by similarity percentage and suggests context-aware substitutes (Peng, 2024; Ruiz-Rincón & Galpin, 2024). Simultaneously, local caching prevents state resets during app switching (Churik, 2025).

Benefits:

- Flexible Ingredient Matching: Accommodates varied ingredient phrasing and partial pantry lists without search failures (Pal, 2023).
- Household Food Waste Reduction: Prioritizes recipes utilizing existing perishable leftovers before they spoil (Woolley et al., 2022).
- High Cross-Device Reliability: Preserves user pantry entries locally during network drops and synchronizes changes seamlessly across devices (Churik, 2025).

Limitations and Risks:

An alternative approach is a Particle Swarm Optimisation (PSO) Metaheuristic Engine, which optimizes weekly meal sequences against ingredient expiry dates and pack sizes to minimize household food waste. However, PSO algorithms carry higher computational complexity and are prone to premature convergence during real-time ad-hoc queries (Woolley et al., 2022). Therefore, the NLP Content-Based Vector Model is chosen because it executes similarity rankings in milliseconds, handles unstructured user text inputs naturally, and scales efficiently for interactive mobile recipe discovery (Pal, 2023; Ruiz-Rincón & Galpin, 2024).

Alternative Approach Comparison:

An alternative approach is a Particle Swarm Optimisation (PSO) Metaheuristic Engine, which optimizes weekly meal sequences against ingredient expiry dates and pack sizes to minimize household food waste (Woolley et al., 2022). However, PSO algorithms carry higher computational complexity and are prone to premature convergence during real-time ad-hoc queries (Woolley et al., 2022). Therefore, the NLP Content-Based Vector Model is chosen because it executes similarity rankings in milliseconds, handles unstructured user text inputs

naturally, and scales efficiently for interactive mobile recipe discovery (Pal, 2023; Ruiz-Rincón & Galpin, 2024).

5.0 Github repository

Github link: https://github.com/bungalilyyy/CSC649-Assignment1-SupercookRedesign

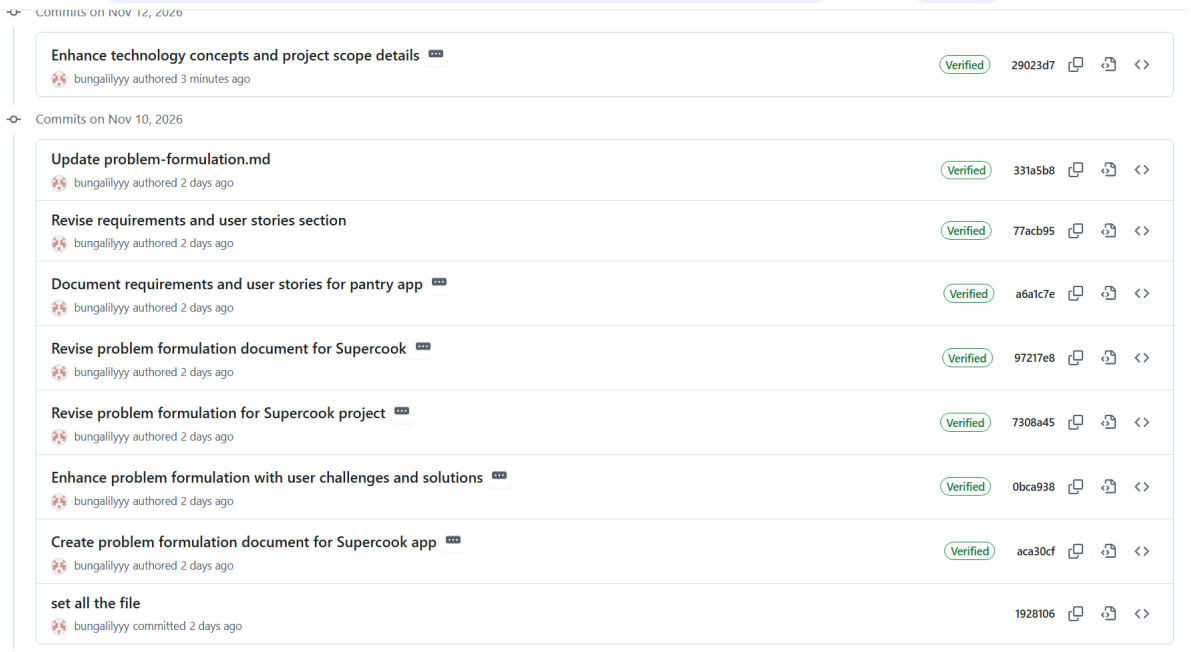


Figure 5.0 commits history



Figure 5.1 folder in github

References

Bill S. (2023, May 18). [Review of the mobile app SuperCook – Recipe Generator]. Google Play Store. https://play.google.com/store/apps/details?id=com.supercook.app

DJ. (2024, December 6). [Review of the mobile app SuperCook – Recipe Generator]. Google Play Store. https://play.google.com/store/apps/details?id=com.supercook.app

ReNew Innovations. (2020, September 14). [Review of the mobile app SuperCook – Recipe Generator]. Google Play Store. https://play.google.com/store/apps/details?id=com.supercook.app

Churik, B. (2025). Data synchronization strategies in cross-platform mobile applications offline. *Sciences of Europe, (174)*, 88–93. https://doi.org/10.5281/zenodo.17335444

Pal, S. (2023). *Recommendation system for food dishes in specific restaurants based on sentiment analysis* [Master's thesis, National College of Ireland]. https://norma.ncirl.ie/7242/

Peng, Y. (2024). *SuperCook: Design interactive culinary experiences with AI technology* [Master's thesis, Rochester Institute of Technology]. RIT Digital Archive. https://repository.rit.edu/theses/11759/

Ruiz-Rincón, S., & Galpin, I. (2024). SnapChef: AI-powered recipe suggestions. In *Proceedings of the Workshops at the 7th International Conference on Applied Informatics 2024* (CEUR Workshop Proceedings, Vol. 3795, pp. 28–40). CEUR-WS.org. https://ceur-ws.org/Vol-3795/icaiw_waai_3.pdf

Woolley, E., Luo, Z., Jellil, A., & Simeone, A. (2022). A data driven approach to reducing household food waste. *Sustainable Production and Consumption, 29*, 600–613. https://doi.org/10.1016/j.spc.2021.11.004

CSC649 — SPECIAL TOPICS IN COMPUTER SCIENCE

Assignment 1: Cover Sheet

Put this page at the front of your PDF report.

Student name	Nur ‘Aliyah Binti T.N Winda Wijaya
Student ID	2025471362
Class	CS2305D
My problem (one line)	Home cooks encounter missing pantry information, disappearing saved recipes, and recommendations containing unavailable ingredients when using recipe applications.
Specialized technology chosen	Natural Language Processing
GitHub profile link	https://github.com/bungalilyyy/CSC649-Assignment1-Supercook Redesign
Project repository link	https://github.com/bungalilyyy/CSC649-Assignment1-Supercook Redesign
Repository status (Public, or lecturer added)	Public
Did you use GenAI? Which tools?	Yes, Notebook and Gemini
FSKM GenAI declaration submitted? (Yes / No)	Yes

Declaration

I confirm that this work is my own. I have checked all facts and references, I have declared my use of GenAI, and I have followed the FSKM GenAI guide and UiTM academic integrity rules.

Signature: _____ Date: _____12/11/2026_____